

Internal Test

Question No. 2

Upload your answer in 30 minutes

2. Brake mechanism used to reduce gun recoil consists of piston attached to barrel moving in fixed cylinder filled with oil. As barrel recoils with initial velocity v_0 , piston moves and oil is forced through orifices in piston, causing piston and cylinder to decelerate at rate proportional to their velocity. Determine $v(t)$, $v(x)$, $x(t)$.

Roughly sketch

$v(t)$ vs. t , $v(x)$ vs. x , and $x(t)$ vs. t .

(10Marks)

